



SECR4483 Secure Programming

GROUP PROJECT – SECURE APPLICATION DEVELOPMENT

Requirement

- Work in groups of **3 or 4 students (not more nor less)**.
 - Integrate work from **Part 1 (Authentication Module)** and **Part 2 (Transaction Module)**, or develop a new secure application based on both components.
 - **Backend:** PHP/Python/Node.js or similar
 - **Frontend:** HTML/CSS/JavaScript
 - **Database:** MySQL or equivalent
 - **Tools:**
 - Vulnerability Scanner (e.g., Burp Suite, OWASP ZAP)
 - Code Analyzer (e.g., SonarQube or manual review)
-

PART 1 – AUTHENTICATION MODULE ANALYSIS (5 MARKS)

Objective:

- Evaluate the security of a login and registration system.

Task:

- Select a previously developed or new login system.
- Analyze the code for:
 - Input validation flaws
 - Weak password handling
 - SQL injection/XSS vulnerabilities
 - Session initialization risks
- Propose theoretical improvements.

Deliverable:

- **Brief report** (1–2 pages) with:
 - Identified vulnerabilities and impacts
 - Recommended improvements
 - **Code snippets** and **screenshots** as evidence

Rubric:

Criteria	Marks
Identification of vulnerabilities	2
Explanation of security implications	1
Suggested improvements	1
Evidence: code and screenshots	1

PART 2 – TRANSACTION & SESSION MODULE ANALYSIS (5 MARKS)

Objective:

- Assess the security of a transactional system with focus on session handling and input protection.

Task:

- Choose a previously developed or new transaction/order module.
- Analyze the code for:
 - Session fixation or hijacking
 - Client-side state manipulation
 - Authorization gaps
 - Input/output sanitization flaws
- Suggest theoretical countermeasures.

Deliverable:

- Brief report** (1–2 pages) with:
 - Identified threats and potential exploits
 - Suggested improvements
 - Code snippets** and **screenshots** as evidence

Rubric:

Criteria	Marks
Identification of threats	2
Risk explanation and impact	1
Suggested security improvements	1
Evidence: code and screenshots	1

PART 3 – FINAL INTEGRATED SECURE APPLICATION (7 MARKS)

Objective:

- Build a fully integrated web application that combines login and transaction features with robust security controls.

Task:

- Merge the findings and features from Parts 1 and 2 into a functional application.
- Implement the following **security elements**:
 - Input validation and output sanitization
 - Multi-factor authentication (optional)
 - Role-based access control
 - Secure session management (e.g., timeout, regeneration)

Rubric:

Criteria	Marks
Functional integration (auth + transaction)	2
Security implementation (at least 4 elements)	3
Testing and evaluation	1
Documentation quality	1

- Encrypted storage of sensitive data
- Secure transmission (e.g., HTTPS or simulated encryption)
- Proper error handling with limited information disclosure
- Conduct vulnerability testing using tools such as Burp Suite or OWASP ZAP.

Deliverables:

- Final **secure web application (source code in ZIP)**
- **Technical report (PDF)** including:
 - Integration summary
 - Implemented security measures
 - Results of security testing (tool output screenshots)
 - Summary of Part 1 & Part 2 findings (appendix)
- **Screenshots** of key features and interface
- **GitHub link** (optional but encouraged)

VIDEO PRESENTATION (3 MARKS)

Objective:

- Demonstrate the functionality and security of the final application.

Content to include:

- System demo (login to transaction flow)
- Key security features walkthrough
- Diagrams (UML, action flow)-Optional
- Highlights from security analysis and testing

Deliverables:

- Upload to **YouTube (set as Unlisted)**
- Put the link at the front page of your technical report

Rubric:

Criteria	Marks
Clarity of demo and workflow	1
Explanation of security features	1
Visuals/screenshots/testing evidence	1

- END -